

When does Label Smoothing Help? Research Report.

The provided research paper published by Google Brain, Toronto goes deep into Label smoothing. It explains 'when and why does label smoothing help' by experimentally studying its core neural structure.

The Problem

Even though label smoothing was widely used, no one knew why it worked. Researchers used it because it improved results. Before this paper, there was no proper study on how label smoothing affected the internal representation of neural network.

Over confident models were another problem that the research paper has explained. Normally trained neural networks are majorly trained using hard targets, i.e. correct class gets a probability of 1 and all other classes get a probability of 0. This makes the model overconfident. It starts ~~given~~ giving outputs like 99% for correct class even when it isn't completely sure. Label smoothing was introduced to tackle this. Instead of using hard ones and zeroes we soften the targets. This method proved to be more efficient in image classification, speech recognition and machine translation as it reduced the problem of overfitting and calibrated the model.

Example (Fashion MNIST Data)

For a T-shirt.

Normally trained model would predict

T shirt	Shirt	Pants	Shoes
0.99	0.006	0.002	0.001

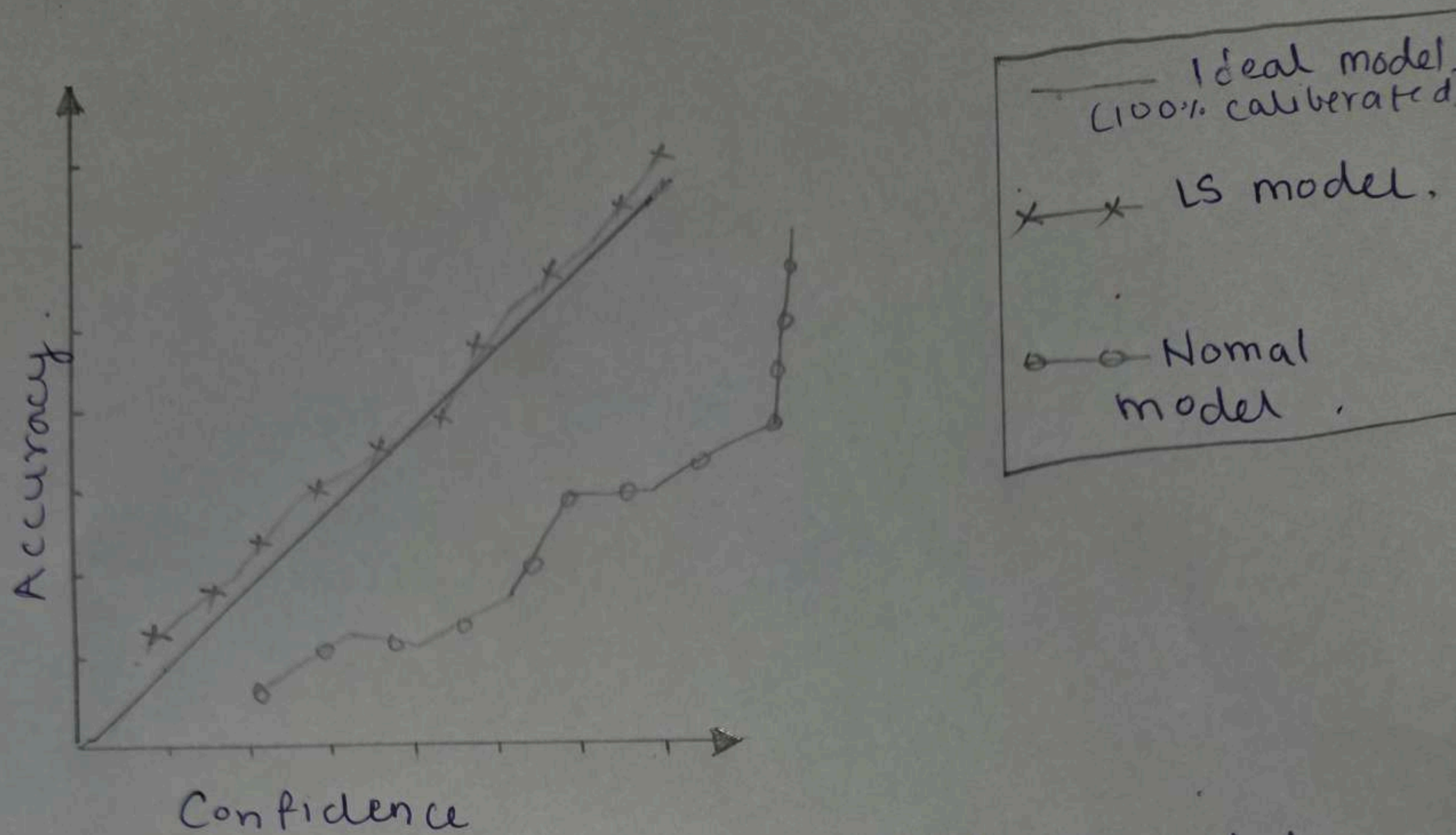
Model Trained w/ LS would Predict ($\alpha=0.1$)

T shirt	Shirt	Pants	Shoes
0.91	0.01	0.01	0.01

A normally trained model is over confident (0.99) while the LS trained model is honest about its prediction (0.91) and gives small but equal probability to other classes instead of near zero.

Why is overconfidence harmful?

Overconfidence can be harmful in many ways. The most direct way is poor calibration. The model ~~says~~ predicts 99% but in reality it might only be correct 80% times. This means the confidence score becomes meaningless. In real life, a model can tell that a patient is 99% not diagnosed but its true only 75% times. This this could prove very harmful. The LS trained models seem to be correct about their confidence. When they predict 85% they are mostly correct $85 \pm 2\%$ of the time.



This graph shows calibration of both models and compares them to ideal model.

Other problem with overconfidence is overfitting. When a model is over confident, it starts memorising values instead of generalising it over the dataset. This decreases its accuracy greatly when implemented on test dataset.

How Label smoothing works

Label smoothing works by modifying the targets. Instead of using hard targets, we soften them by distributing small amount of target to all classes.

The formula used is

$$y_{(LS)} = y(1-\alpha) + \alpha/k$$

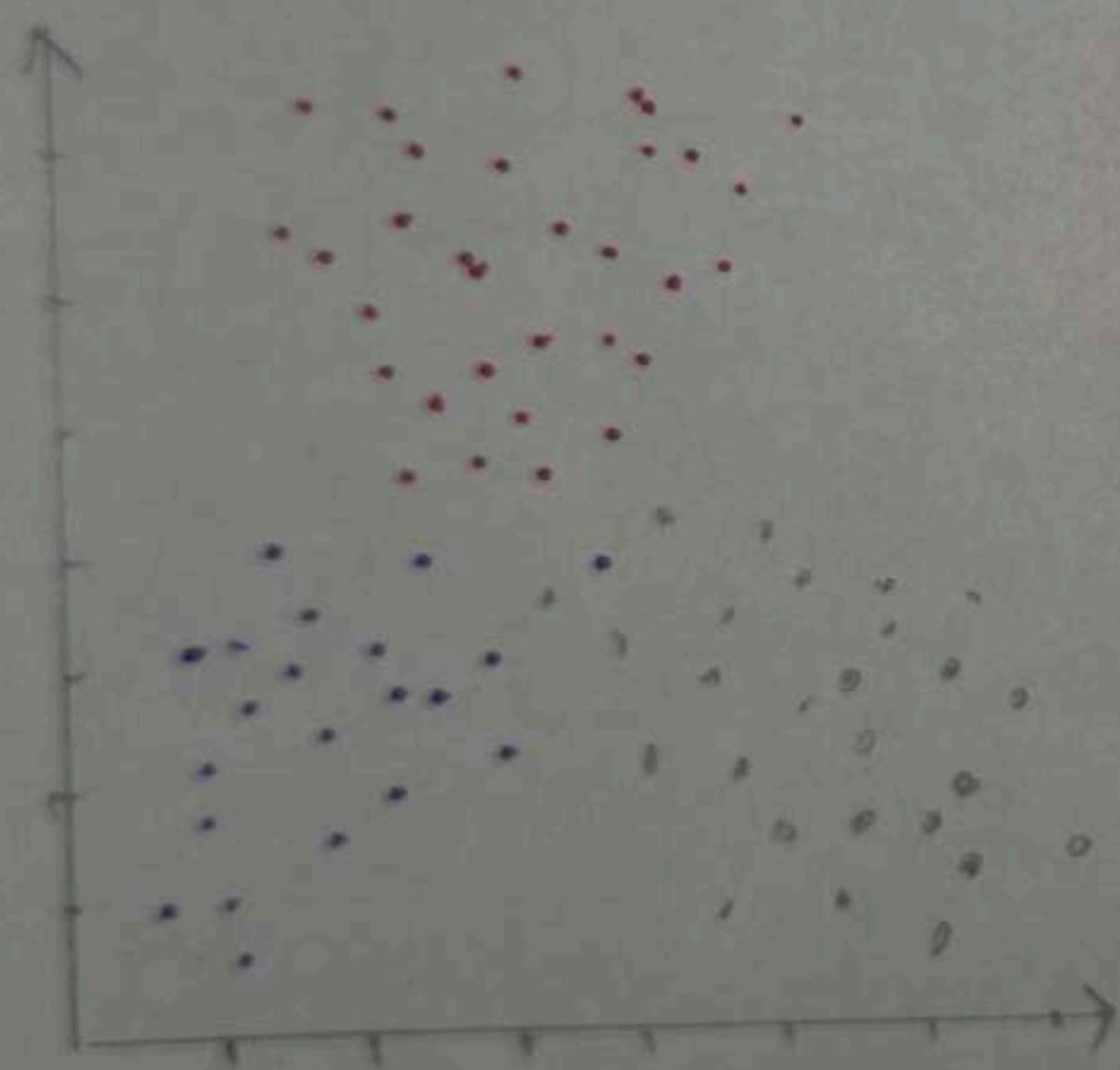
where y is the original hard target, α is smoothing parameter and k is number of classes.

So fashion MNIST Dataset with $\alpha=0.1$ & $K=10$.

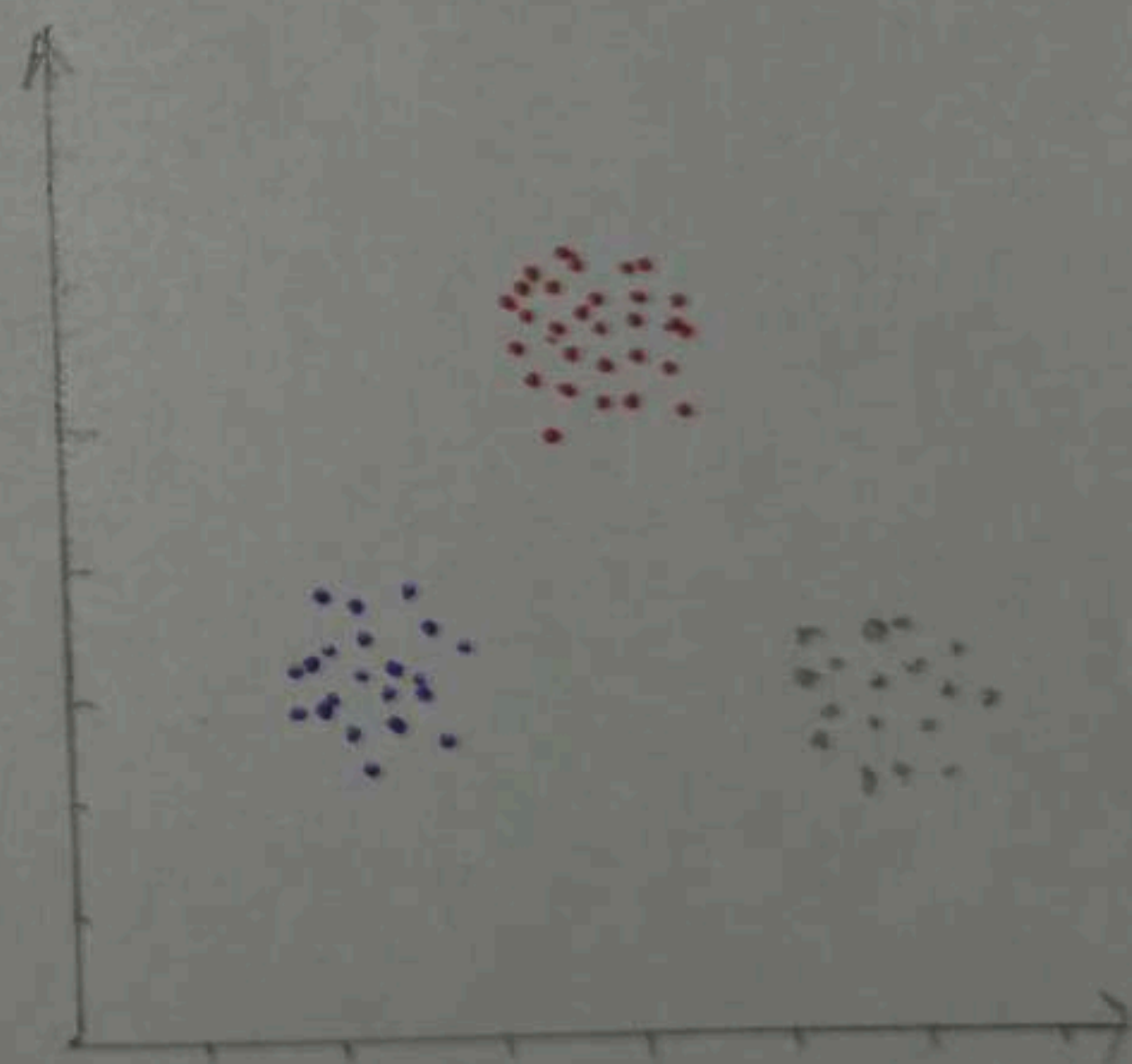
- Correct class gets: $1 \times 0.9 + 0.01 = 0.91$
- Wrong class gets: $0 \times 0.9 + 0.01 = 0.01$.

This way, the model never pushes probability to extremity, instead it is trained to get moderate confident values. This prevents weights from growing large and stops overfitting.

The paper shows that this causes the penultimate neuron layer to form tight equally spaced clusters as probability of correct class is nearly equal and every wrong class is equally spaced.



Training w/o LS



Training w/ LS

This geometric structure is what gives label smoothing its generalisation and calibration benefits

Advantages of Label Smoothing

The most direct advantage is improved generalisation. Preventing the model from being overconfident and stopping overfitting. The model performs better on unseen data with label smoothing.

The second advantage is better calibration. This way model's confidence actually reflects how accurate it is.

Third advantage is that the parameters are not pushed to extremes, this improves smoothness and stability of training to some extent.

Due to all this, label smoothing has shown improved results for ~~many~~ various tasks.

Limitations of Label Smoothing

Most significant limitation is that label smoothing hurts knowledge distillation. When a teacher model is trained with LS, it becomes a poor teacher.

This is because LS forces all wrong classes to have equal probability which erases similarity in between classes.

Other limitation is that high alpha value hurts accuracy. It is seen in my experiments, as we increase the α , accuracy drops significantly thus destroying model's confidence. Typically $\alpha=0.1$ is best.

LS trained model assumes uniform distribution among wrong classes, for it, a t-shirt is as similar to shoes as it is to a skirt.

Future improvements and extension

- Information Bottleneck - Good generalisation comes from compressing info. LS does exactly this. It compresses info into tight clusters. The paper suggest exploring this connection deeply.
- Distillation of translation - The google brain team only studied distillation for image classification. Translation is still to be studied.
- Can we design something that gives calibration and generalisation benefits without hurting distillation is a big question I am thinking of.

Summary.

- Problem → overconfident models, LS used blindly and not studied.
- overconfidence → overfitting, poor calibration.
- Working → formula, clustering, calibration effect.
- Advantages → generalisation, calibration, non-extreme weights.
- Limitations → distillation hurts, erases b/w class info.
- Future → Information bottleneck connection, better distillation method.